

# Multi-Scale Modelling of Biological Systems

## Assignment 4

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September 2026

### 1 Introduction

In this study we investigated whether different models (waveform,uninspired and inspired) accurately represent real brain activity in the STG region of the brain. Using representational similarity analysis (RSA), we compared neural responses to 288 sounds and their relevant STG fMRI responses with 3 different architectures and pretrained YAMNet. By comparing trained and untrained networks, we examine how learning changes brain alignment. We also investigate whether alignment varies with architectural complexity and network depth.

Our purpose for this assignment was to determine whether training and architectural choices produce representations more similar to actual measurements in the brain.

Our study was primarily a contrastive study as we compared the correlation from RSA for 3 different models with the STG audio data we have. The added model would add a constructive element to our study

#### 1.1 Data

We used 288 sounds from the Santoro dataset and their YAMNet embeddings alongside the STG fMRI responses.

#### 1.2 Models

We used 3 pretrained YAMNet, WaveformModel, UninspiredModel and InspiredModel, The three models were trained on ESC-50 Dataset.

## 2 Methods

For each of the models we constructed a representational dissimilarity matrix (RDM) of euclidean distances between sound representations. We followed the same procedure for the STG response patterns. We then calculated the Pearson correlation between the upper-triangular part of the model RDMs and the brain RDMs while excluding the diagonal. We report all subsequent analyses as correlation between the model’s upper-triangular matrix entries, averaged across layers, and the brain responses for each individual.

The alignment was calculated as Pearson correlation between upper triangular of the matrices and the brain for descriptive purposes we decided to report layerwise mean correlations( $r$ ) and best layer correlation and changes in correlation if present in the model. Positive values would imply that sounds that are more dissimilar according to the model tend to also be more dissimilar in the STG responses. It is important to note that these are not measures of variance explained but rather similarity of representational dissimilarities.

Effects of training were then calculated as changes in the mean pearson correlation.

We visualized STG responses using t-SNE.Flattened representations were reduced to at most 50 PCA components. This process was then followed by computing two dimensional t-SNE with perplexity 30, PCA initialization, automatic learning rate, 1,000 iterations, seed 42. Points were colored according to the datasets 73 sound categories. Embeddings were independently fitted and interpreted through grouping local categories; axes and between-cluster distances were not compared.

## 3 Results

### 3.1 RDM Comparison

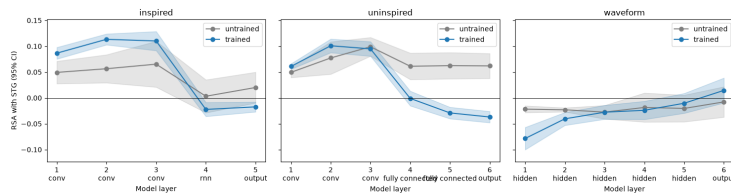


Figure 1: Comparison of RDM Mean Correlations

**Fig. 1** Trained version of InspiredModel and UninspiredModel showed their strongest alignment in the convolutional layers, highest value at  $r = .11$  and  $r = .10$ , respectively. Alignment as a result, declined, becoming negative in the recurrent or later fully connected representations. On the otherhand, Waveform

model showed increasing alignment with depth, although its scores remained comparatively low..

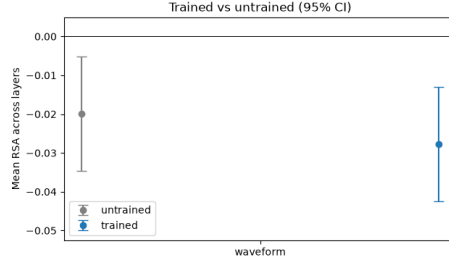


Figure 2: Trained vs Untrained Waveform

**Fig. 2** The results for Trained Waveform model show that alignment increased with depth, from approximately  $r = -.08$  at the first hidden layer to  $r = .015$  at the output layer. The untrained Waveform model remained slightly negative

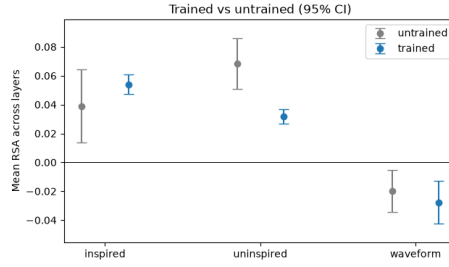


Figure 3: Comparison between trained and untrained versions of all three models

**Fig. 3** Among the trained models, InspiredModel achieved the highest mean layerwise RSA (.054), which was followed by UninspiredModel (.032) and WaveformModel (-.028); all pairwise differences were statistically significant after Holm correction ( $p < .001$ ). After training, UninspiredModel's showed reduced alignment ( $p = .010$ ), while at the meantime InspiredModel's increase was not significant ( $p = .347$ ).

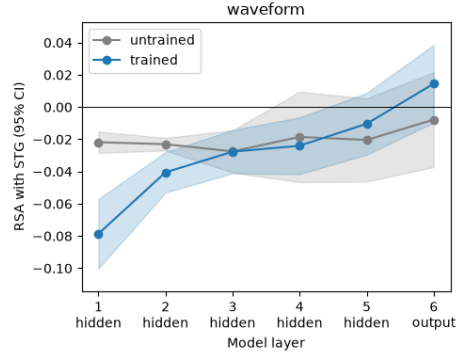


Figure 4: Waveform RSA with STG 95 percent CI

**Fig. 4** WaveformModel’s mean layerwise RSA decreased from  $-0.020$  before training to  $-0.028$  after training. Both means were slightly negative, and the training effect was not statistically significant (Holm-adjusted  $p = .347$ ). Error bars show 95% confidence intervals across five runs.

### 3.2 T-NSE

**Fig. 5** The t-SNE plots suggest that training reorganized sound representations, producing more distinct local groups in some models. However, categories remained mixed, and clearer grouping did not consistently accompany stronger RSA. These visualizations therefore help interpret representational changes but do not establish brain alignment, particularly because they show selected layers from one run and were fitted independently.



Figure 5: t-SNE Plot

## 4 Discussion

The RSA results show that brain alignment depends on both architecture and model depth. The InspiredModel showed the highest trained mean RSA, but its best alignment occurred in the convolutional layers, while the recurrent and classifier layers showed weak or negative correlations. A similar pattern was noticed in the UninspiredModel, while the WaveformModel remains the weakest.

In fact, training did not sufficiently improve the results. Some convolutional layers became more similar to STG after training, but later layers often became less aligned. YAMNet showed a similar depth-dependent effect. Early and intermediate layers have positive RSA, while last layers and the final embedding showed little or negative alignment. The t-SNE plots showed changes in category organization after training, but clearer visual clustering did not always correspond to higher RSA. Therefore t-SNE is useful for describing local structure, but RSA remains the more direct measure of brain-model representational similarity.

## 5 Future Work

Future work should test models with more explicit neuroanatomical constraints. For example, separate network stages could represent different auditory regions, with recurrent, lateral, or feedback connections added according to known auditory pathways. Training could also be improved by using larger and more diverse audio dataset.

The RSA analysis also can be repeated with alternative distance and different similarity metrics to test whether the observed model ranking is robust. Finally, data from multiple participants and additional auditory regions would be needed to determine whether the observed preference for intermediate representations generalizes beyond the current dataset.